~~Development Team Project: Project Report~~Airbnb listing prices in New York City: a regression and clustering analysis~~How is distance from Times Square associated with Airbnb listing prices in New York City, and what listing segments can be identified based on location and property characteristics?~~

Development Team Project – Project Report

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(Team F, Group 4)

Introduction

Airbnb provides short-term accommodation across many locations and price ranges in New York City. Location is ~~likely to be~~ an important factor in ~~Airbnb~~ pricing (Deboosere et al., 2019). Therefore, we assume that distance from the most centrally located touristic area is associated with listing price. Times Square has been described as the "epicenter of NYC tourism" (Weaver, 2025).

~~To test this hypothesis, we~~ added a "distance from Times Square in kilometers" field to~~used the 2019 New York City Airbnb dataset~~., ~~to which we added a field to identify the distance from Times Square in kilometres~~. ~~We used regression and clustering to identify different types of Airbnb listings based on~~ distance from Times Square~~location and property characteristics~~. Thus, our research question is as follows:

How is distance from Times Square associated with Airbnb listing prices in New York City, and what listing segments can be identified based on location and property characteristics?~~. Thus, our research question is as follows:~~

~~*How is distance from Times Square associated with Airbnb listing prices in New York City, and what listing segments can be identified based on location and property characteristics?*~~

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We used regression and clustering to identify different types of Airbnb listings. The main purpose of this research is to generate insights that could help Airbnb understand pricing patterns and



different listing segments in New York City.

~~**Figure 1: Map of the five boroughs of New York City**~~
~~*Source: Adapted from Ontheworldmap.com team (no date)*~~

Data Preparation

We ~~first~~ cleaned the original 2019 Airbnb New York dataset ~~that had~~ (48,895 Airbnb listings). The ~~final~~ cleaned version ~~dataset~~ consisted of 48,884 listings. ~~As a final step, we created the~~We added an extra column - distance_times_square_km, ~~column~~ which calculates distance from Times Square using latitude and longitude.

Exploratory Data Analysis

We ~~first~~ first visualised listing prices according to geographic location~~:. This resulted in the following map:~~

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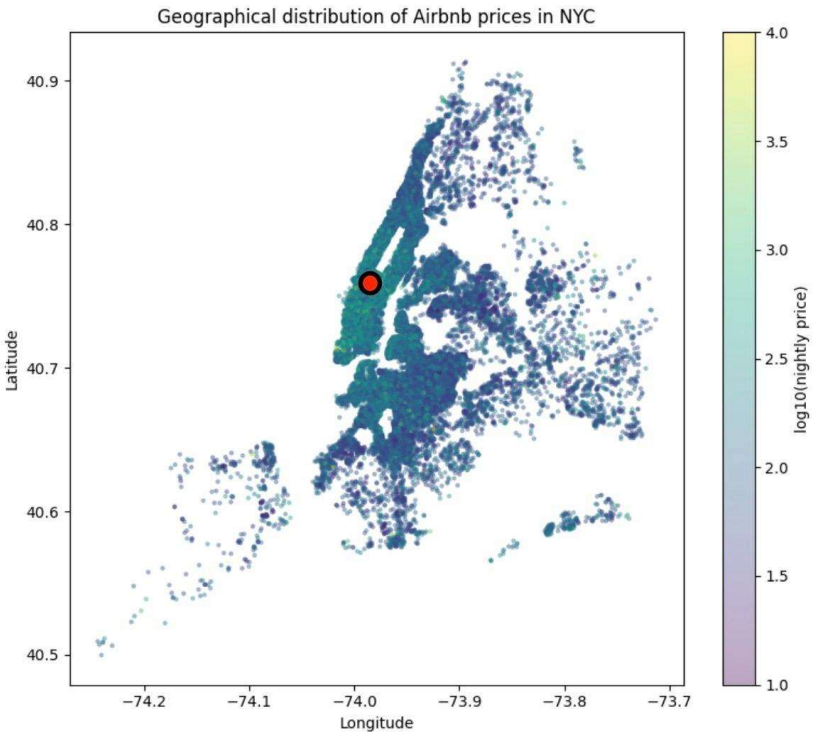


Figure 2: Geographical distribution of Airbnb prices in NYC

~~*Source: Team F analysis of the AB_NYC_2019 dataset (2026)*~~

The red dot represents the location of Times Square. ~~H~~We ~~can clearly see that h~~igher prices are concentrated around Manhattan/Times Square. Listings further away are generally cheaper.

Next, we visualised the median Airbnb price by distance from Times Square:

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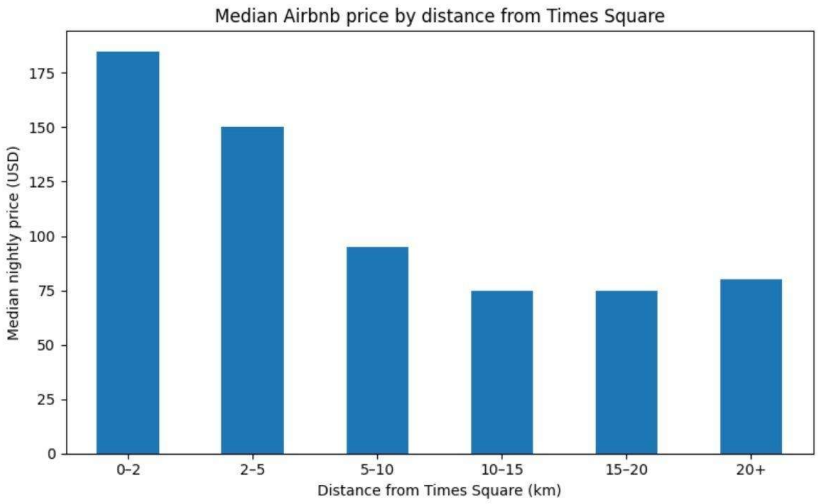


Figure 3: Median Airbnb price by distance from Times Square
Source: Team F analysis of the AB_NYC_2019 dataset (2026)

Listings within 0-2 km of Times Square have a median price of $185. At 2-5 km, this falls to $150. At 5-10 km, it fallsgoes falls even lower to $95.For a distance >10km If the property is further away than 10 km, the price is around $75.

Evidently, The pattern, therefore, is that Airbnb prices tend to decrease as distance from Times Square increases. Interestingly, Since the prices slightly increase s again from 20+ km, the making the relationship is not perfectly straightlinear straight.

However, distanceHowever, distanceDistance from Times Square might not be the only factor influencing reason prices gradually decrease. BListings closer to Times Square may also be different because of borough, room type, quality, neighbourhood and other characteristics may also have an influence. Therefore, regression and clustering are used to further investigate these relationships. in more detail.

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Regression

Through regression, we predicted~~Framing our question as a regression problem means finding a model that predicts~~ the value of price based on how one or more variables change. ~~W~~To do so, ~~w~~e firstly decided to transform price into its logarithmic form log(price) to reduce the impact of ~~the few listings that have an~~ outlying price~~s~~ (5-10k £).

We checked multicollinearity amongst Distance and Borough ~~to see whether it would be difficult for the model to separate their individual effects on price. Multicollinearity is when two or more variables have a high correlation, meaning it's difficult to accurately establish their individual effects (Wooldridge, 2020)~~. Numbers greater than 10 could represent severe multicollinearity, whilst lower numbers show reduced to no multicollinearity (Wooldridge, 2020). ~~We checked multicollinearity amongst Distance and Borough to see whether it would be difficult for the model to separate their individual effects on price~~: Distance was not problematic (Variance Inflation Factor (VIF) = 1.9). However, Borough analysis showed a VIF >10, which is high. Thus, ~~This warned us that~~ the split of credit between Borough and Distance should not be taken as separate causal effects.

~~Secondly, the data was split~~Secondly, w~~W~~e split the data into a test and a training set, respectively containing 20% and 80% of the data. We used Borough to stratify the split so as to ~~To do so we also stratified by borough to~~ avoid underrepresenting the rarest labels. ~~A stratified split involves~~ and ensuring the data splits each contain proportionate amounts of the overall data (Géron, 2025)

We ~~then~~ used three different regression approaches and compared them:

1. A simple linear regression ~~(where one item is used to predict an outcome)~~ on distance alone found each +1km was associated with a ~5.9% price drop (R^2=0.15).



Figure 4: Log(price) vs. distance

~~Source: Team F analysis of the AB_NYC_2019 dataset (2026)~~

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2. A multilinear regression (where two or more items are used to predict an outcome) approach with all features; controlling for room type and borough caused distance's effect to shrink, showing itthat part of its raw association in the simple linear model was probably an effect mediated by borough and room type.

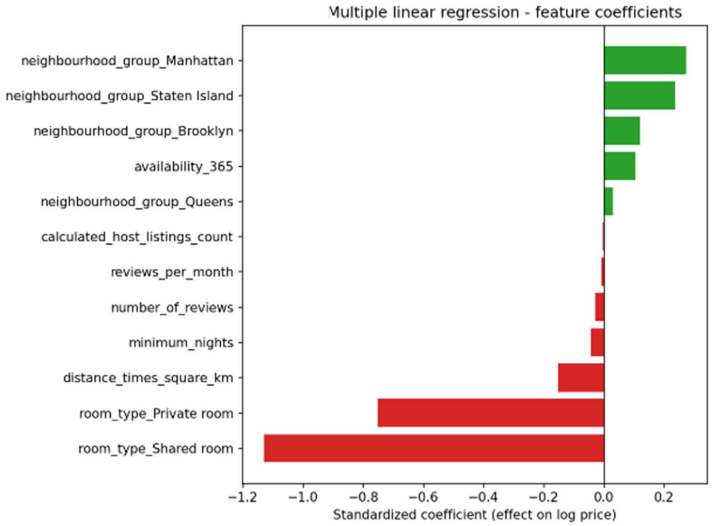
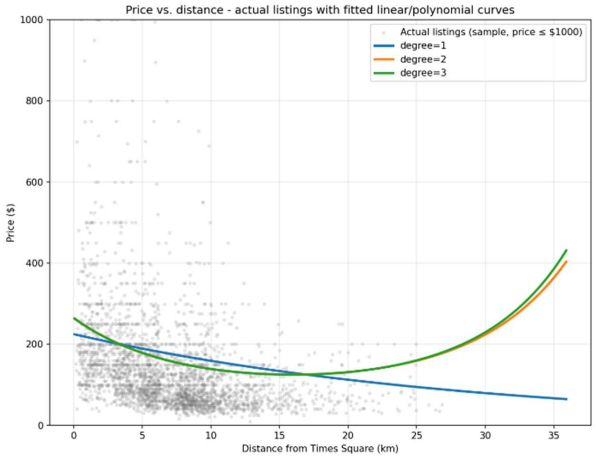


Figure 5: Multiple linear regression - feature coefficients

Source: Team F analysis of the AB_NYC_2019 dataset (2026)

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3. Polynomial regression is a modelling of the relationship between a dependent variable and one or more independent variables, allowing nonlinear relationships to be represented. (Géron, 2025) We also performed a polynomial regression approach to determine nonlinear relationships (Géron, 2025).In performing polynomial regression, Wwe checked first with degree 2, which improved fit (R^2=0.51), revealing price drops



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steeply near the centre and then a flattening outflattens further out. Degree 3 didn't add any significant improvement. In the figure, degree 1 is the linear model.

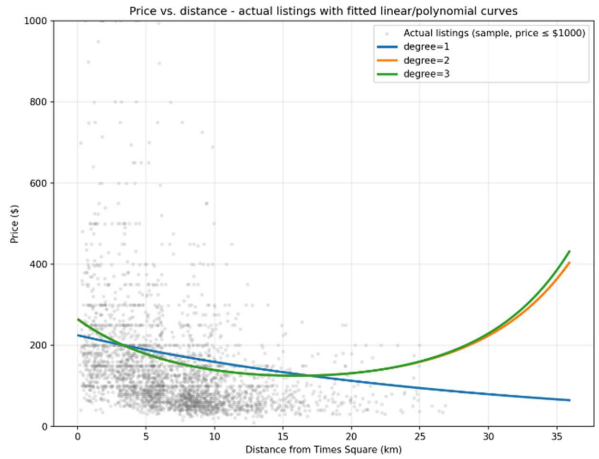


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The degree-2 polynomial achieved a test RMSE of 0.488 (log-price units) and R^2 of 0.512, both an improvement over the plain linear model (RMSE 0.496, R^2 0.496). This confirms that the true relationship between distance and price is not linear.

Ultimately, the learning curves showed train/validation RMSE converging closely. , ~~which told us~~This shows that the model fairly predicts price, without overfitting the data~~. Overfitting is where~~

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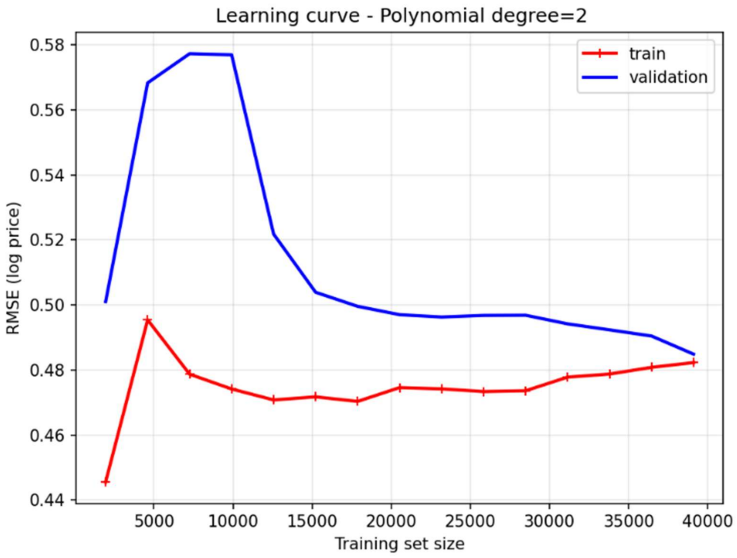
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~~the model is so complex that it extracts random correlations from the training data that don't fit the actual data.~~ (Géron, 2025).

Figure 7: XXXX XXXX XXXX XXXX XXXX



There were no signs of overfitting, so we chose not to use any regularisation technique on the model.

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Clustering

While regression examined the relationship between listing price, distance and other characteristics, clusteringClustering was used to identify distinct groups of Airbnb listings with similar location and property characteristics, providing a further complementary perspective on the patterns within the dataset. **[REF: clustering/unsupervised learning]**.Clustering is the practice of grouping similar items together to identify both groups and outliers. Géron (2025)

The optimal number of clusters was evaluated using the elbow method – identifying the point at which the data turns a corner and no longer grows (Géron (2025) - [REF] and average silhouette scores. **[REF]**. Although k = 2 produced the highest silhouette score (0.392 - from a range of -1 to +1 (Géron 2025)), this solution was considered too coarse to provide meaningful segmentation beyond the existing price- distance findings. The three-cluster solution achieved the next highest substantively meaningful silhouette score (0.252) and corresponded with the point at which improvements in within-cluster variation began to flatten; therefore, k = 3 was selected.

K-means clustering (an unsupervised learning algorithm that clusters iteratively (Géron 2025)) **[REF]** was performed on six scaled variables. [REF: standardisation/feature scaling].Min-max scaling and standardisation techniques as described by Géron (2025) were used. Figure 8X uses principal component analysis (PCA), which rotates the data in a way that makes it easier for decision trees to split data (Géron 2025). **[REF: principal component analysis (PCA)]** to visualise the resulting clusters in two dimensions.; Cclustering was not performed on the principal components themselves. Together, PC1 and PC2 explain 52.5% of

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the variance. The clusters occupy distinguishable regions of the PCA space, although some overlap remains.



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~~Figure X: PCA projection of K-means Airbnb listing clusters (k = 3)~~

~~*(Source: Team F analysis of AB_NYC_2019 dataset using R.)*~~

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Figure 8: PCA projection of K-means Airbnb listing clusters (k = 3)

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Cluster profiling (Table 1) indicates that Cluster 1 represents a ~~smaller professional/~~long-stay segment, ~~with a characterised by a~~ 30-night median minimum stay, high availability (316 days) and hosts with a median of 16 listings. Cluster 2 represents ~~active value~~ short-stay listings, with the lowest median price ($95), greatest distance from Times Square (8.7 km) and highest review activity (2.12 reviews/month). Cluster 3 comprises the largest segment (54.6%), characterised by zero median availability and low review activity.

~~Table 1: Profile Characteristics of the Three K-Means Airbnb Listing Clusters~~

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	Cluster 1	Cluster 2	Cluster 3
Size	4,594 (9.4%)	17,608 (36.0%)	26,682 (54.6%)
Median price	$150	$95	$115
Median distance from Times Sq.	3.9 km (closest)	8.7 km (furthest)	5.7 km (middle)
Median minimum nights	**30**	2	3
Median # reviews	1	28	2
Median reviews/month	0.09	2.12	0.12
Median availability (days/365)	**316**	197	**0**
Median host listings count	**16**	1	1
Dominant room type	Entire home/apt (68%)	Private room (54%)	Entire home/apt (55%)
Dominant borough	Manhattan (70%)	Brooklyn (44%), spread across boroughs	Manhattan (49%) / Brooklyn (43%)

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Similarly to regression, ~~Overall,~~ cluster prices decline with increasing distance from Times Square ($150, $115 and $95), ~~supporting the broad regression pattern~~ but ~~while~~ ~~demonstrating that~~ listing segments also differ ~~substantially~~ in host, availability and booking characteristics. However, the moderate silhouette score and overlap in the PCA projection indicate that these segments should be interpreted as broad listing profiles rather than sharply separated groups.

Conclusion

~~Our analysis shows that while properties closer to Times Square can generally be said to have higher prices. However, there is a lot of variability that can only be explained by other factors e.g. room type, availability and neighbourhood rather being purely based on a central location.~~many other factors also play a part in the property's pricing.

Listings further from Times Square tend to be cheaper, although the relationship isn't perfectly linear. But borough, room type and other listing characteristics also explain a large part of price differences. The clustering backed this up, revealing three broad types of listings with different prices, availability, host behaviour and review activity. Overall, Airbnb pricing reflects a mix of location and listing characteristics, rather than any single factor.

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2019 Airbnb New York dataset

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